

Project 1 report: Music Carousel

The Romantic Tour group

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Project description

Music Carousel is a console-based playlist manager. Songs are stored in a circular doubly linked list, where a cursor tracks the currently playing track separately from the list's display order. Users can move forward and backward through the playlist, jump to a specific track, add or remove songs, queue a song to play next, and shuffle the playlist while keeping its total duration unchanged.

The program also keeps a short history of the last five played songs, displays the total playlist duration, and supports two repeat modes: repeat the whole playlist, or repeat the current track. Playlists are loaded from and saved back to a text file, so changes persist between runs.

Check the codebase at: [GitHub](#)
More about the project: [README](#)

Linked list technical decision explanation

We chose a doubly linked circular list because it matches how a playlist actually works. A playlist does not really have an end. When the last song finishes, it goes back to the first one, and a circular list does that on its own: advancing from the last node lands you on the first. That means our `next()` function never has to check whether it reached the end, while a linear list would need that check every single time.

We made the list doubly linked so that going backward works the same way as going forward. In a singly linked circular list, finding the previous track means walking $n-1$ nodes to get around to it, which is $O(n)$. With prev pointers it is one step, so $O(1)$. Those pointers also make editing the playlist cheap. Because we already have the node we are on, adding a song right after the current track or removing the track that is playing both take **constant time**. And since each node is allocated on its own, the playlist can grow one song at a time without copying everything over

Instructions:

Instructions to run:

```
g++ -std=c++17 -Wall -Wextra main.cpp Song.cpp Playlist.cpp -o player  
./player
```

Note: playlist.txt must be in the same directory as the executable.

Instructions to interact with program:

The program displays the playlist with track numbers, durations, and a marker showing the current track. Enter a single letter at the prompt to choose an action:

- n - play the next track
- p - play the previous track
- g - jump to a track by entering its number
- r - toggle repeat mode (all/one)
- a - add a song to the end of the playlist (enter title, artist, duration)
- q - queue a song to play immediately next
- x - remove the current song
- s - shuffle the playlist
- w - save the playlist back to playlist.txt
- e - exit the program

Roles

Minh Duong Nguyen (U80322005): 50%	Ngoc Viet Nguyen (U14177860): 50%
Designed and built the core circular linked list architecture	Implemented the main playlist application layer: song data, playback logic, and user interface
Wrote the PDF report	Cleaned up codebase

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